

BLISS-SYS-NEGREG-001

Bliss Bot Chapel

system test report

Negative testing, bounded stress testing, regression, and the later PostgreSQL, OIDC, and data-protection acceptance pass.

Item	Value
Application	Bliss Bot Chapel — API, operations console, and public site
Runtime	Linux, .NET SDK 8.0.425, ASP.NET Core
First pass	28 September 2026, no PostgreSQL connection
Acceptance pass	29 September 2026, local PostgreSQL 16 database bliss_phase1
Repository changes	None. Credentials were not committed.

Overall result

The automated regression suite passed all 172 tests on both days. The first pass proved failure handling, rate limiting, and UI resilience while the database was unavailable. The second pass applied the full migration chain, exercised seeded workflows on PostgreSQL, loaded the operations dashboard without HTTP 500 responses, and verified a local OIDC acceptance login with a persistent data-protection key.

Suite	Result
Regression, 28 and 29 September	172 passed, 0 failed, 0 skipped
Negative probes	Expected 400, 401, 403, 404, 405, 429, and 503 responses
Bounded stress, no database	3,000 successful reads; invalid writes throttled
PostgreSQL readiness	Healthy — database connection succeeded
Migration chain	17 of 17 migrations applied
Seeded PostgreSQL workflows	Create, replay, and expected rejections passed
Database-backed reads	600 of 600 returned HTTP 200
OIDC acceptance	Passed when the role claim is in the ID token
Data-protection key	The same key restored the session after restart

1. Regression

Command: `dotnet test BlissBotChapel.sln --configuration Release`. The follow-up result file is `test-results/regression-postgres-followup.trx`. Both executions reported a four-second test duration.

Area	Passed
API, authentication, exports, and frontend routes	69

Area	Passed
Persistence and workflow boundaries	67
Architecture and EF constraints	21
Domain rules and defaults	15
Total	172

Coverage includes creator ingestion, match formation, human review, campaign placement, Wedding Planner isolation, economics phases 1–9, audit export integrity, verification history, runtime hardening, and static frontend routes. The automated suite uses the repository’s in-memory test database and does not replace the PostgreSQL pass below.

2. First pass — no database

This pass started the API in Development with an empty connection string. It established the failure and performance baseline before PostgreSQL was configured.

Negative behavior

Probe	Result
/health/ready with no database	HTTP 503, database unhealthy
Unknown route	HTTP 404
Malformed JSON write	HTTP 400
60 concurrent invalid writes	29 × HTTP 400, then 31 × HTTP 429
Throttled response	Retry-After: 60, correlation ID, and security headers
Production startup without OIDC	Process rejected, exit 134
OIDC enabled without authority and client ID	Process rejected, exit 134
Non-positive write-rate limit	Process rejected, exit 134
Login while OIDC is disabled	HTTP 400, OIDC authentication is not enabled
Request ID not-a-guid	Replaced with a generated GUID
POST to read-only economics observations	HTTP 405
Audit body over 512 KB	HTTP 400, pack exceeds the 512 KB limit
Invalid operations route	Fell back to the overview
Favicon request	HTTP 404 — minor defect

Bounded stress

Scenario	Load	Result	Throughput	p95	Max
Liveness	2,000 / 100	2,000 × 200	3,636.1/s	40.59 ms	80.41 ms
Public page	1,000 / 50	1,000 × 200	3,877.5/s	21.71 ms	33.84 ms
Invalid writes	60 / 30	29 × 400, 31 × 429	2,262.4/s	12.00 ms	14.36 ms

After the write burst, /health/live returned HTTP 200. The operations console stayed navigable while database calls failed. Status showed Process Healthy and Database Unhealthy. These figures are single-host development measurements, not

capacity claims.

3. PostgreSQL acceptance

An isolated local PostgreSQL 16 database named `bliss_phase1` was created. No hosted or production database was used. `dotnet ef database update` applied all 17 migrations:

- 20260919013044_Phase1Foundation
- 20260920063520_Phase2RuleDocument
- 20260920073126_Phase3EvaluationAudit
- 20260920102946_Phase4CreatorIngestion
- 20260920123653_Phase5MatchFormation
- 20260920130043_Phase6HumanReview
- 20260920133028_Phase7CampaignPlacementBinding
- 20260922052555_WeddingPlannerPhase1Foundation
- 20260923045008_EconomicsPhase1Foundation
- 20260923050745_EconomicsPhase2CreatorSnapshots
- 20260923054446_EconomicsPhase3MarketProfiles
- 20260923061423_EconomicsPhase4Recommendations
- 20260923064840_EconomicsPhase5Quotes
- 20260923084506_EconomicsPhase6Compensation
- 20260923100111_EconomicsPhase7ResearchOrchestration
- 20260923104329_EconomicsPhase8WeddingPlannerHandshake
- 20260923115315_EconomicsPhase9HistoricalLearning

Development startup seeded the fictional Phase 1–3, Wedding Planner, and economics reference data. `/health/ready` returned HTTP 200. Every operations dashboard read returned HTTP 200. Counts moved from 7 matches and 3 creators to 8 matches and 4 creators after the workflow writes.

Seeded workflows

Workflow	Result
Creator ingestion create / exact replay	HTTP 201, then 200 with the same run and creator
Negative followers	HTTP 400 — followers cannot be negative
Match formation and replay	HTTP 201, then 200; status APPROVED; same match
Unknown creator	HTTP 400 — creator does not exist
Review of the seeded review-required match	HTTP 201, then 200; result APPROVED
Approval of a non-reviewable match	HTTP 400 — only REVIEW_REQUIRED matches
Placement of the seeded approved match	HTTP 201, then 200; outcome PLANNED
Placement of an unapproved match	HTTP 400 — only APPROVED matches

Final PostgreSQL counts were 4 creators, 8 matches, 1 ingestion run, 1 formation run, 1 review decision, and 1 placement run. The operations ledger showed the creator observation, evaluation, match formation, review, and planned placement.

Database-backed read load

600 mixed reads at 30 concurrent clients covered readiness, creators, matches, campaigns, economics observations, and evaluation runs.

Result	Throughput	Average	p95	p99	Max
600 × HTTP 200	401.1 req/s	70.13 ms	148.65 ms	591.89 ms	637.93 ms

/health/ready remained Healthy after the load.

4. OIDC and data protection

The acceptance process ran as Production on HTTPS, with authentication enabled and a local authorization-code plus PKCE identity provider. The client secret stayed in the process environment.

Check	Result
Anonymous API read	HTTP 401
SSO gate	Continue with SSO; OIDC session required
Completed login	Acceptance Operator
ID-token role bliss.admin	canWrite and canReview true
Invalid write with CSRF token	HTTP 400 validation error
Missing CSRF token	HTTP 403
Login starts	10 × HTTP 302, then HTTP 429
Session after process restart	Still authenticated; creators returned HTTP 200

A roles value supplied only by the userinfo endpoint authenticated the name and email but left roles empty, so writes stayed forbidden. The same role was honored when the identity provider included it in the ID token. An acceptance provider must issue the configured roles claim in the ID token.

The data-protection directory was outside the repository. The API wrote key-c0eca814-abd1-4aca-b4f7-3ef8a7bf0d7e.xml, reused that file after restart, and the status page showed Key ring Configured. The key is persisted without an XML encryptor, so it is not encrypted at rest. The directory must remain access-controlled.

5. Screenshots

First pass — database unavailable

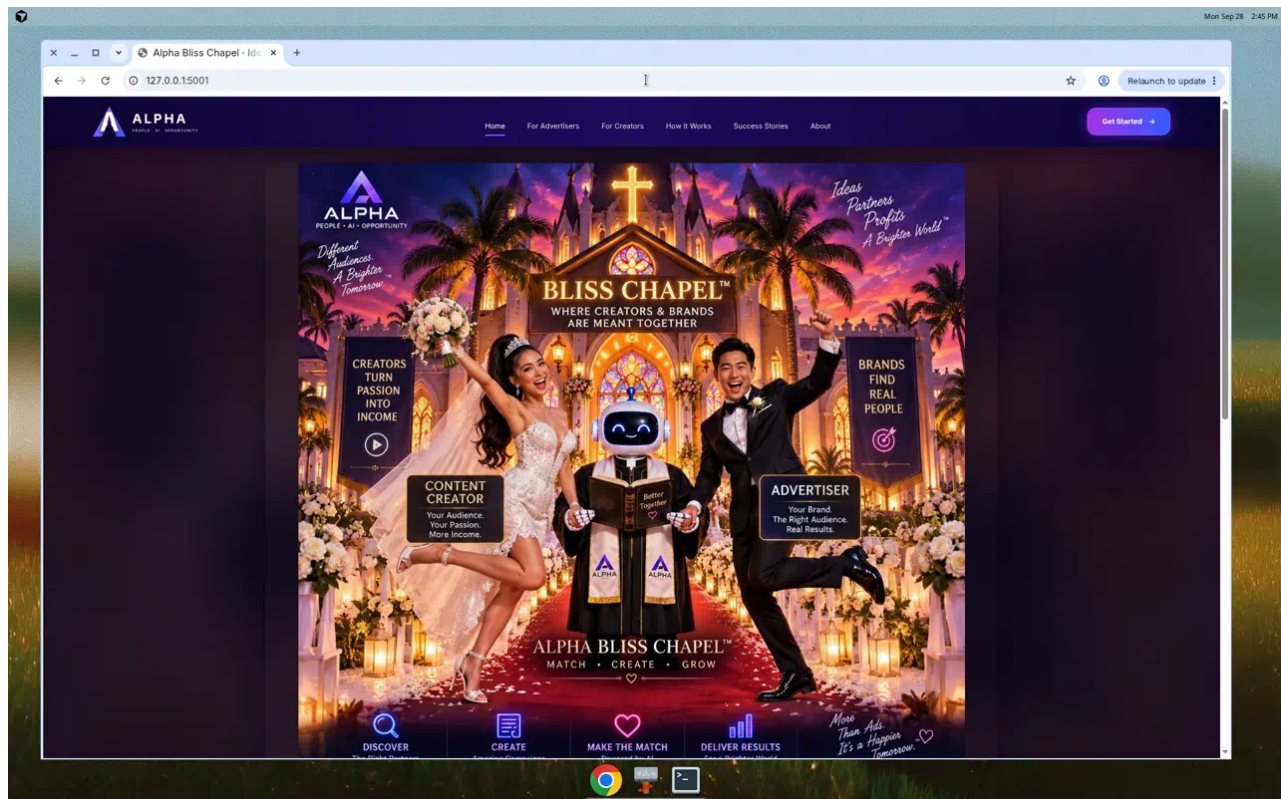


Figure 1. Public homepage rendered while the API had no database connection.

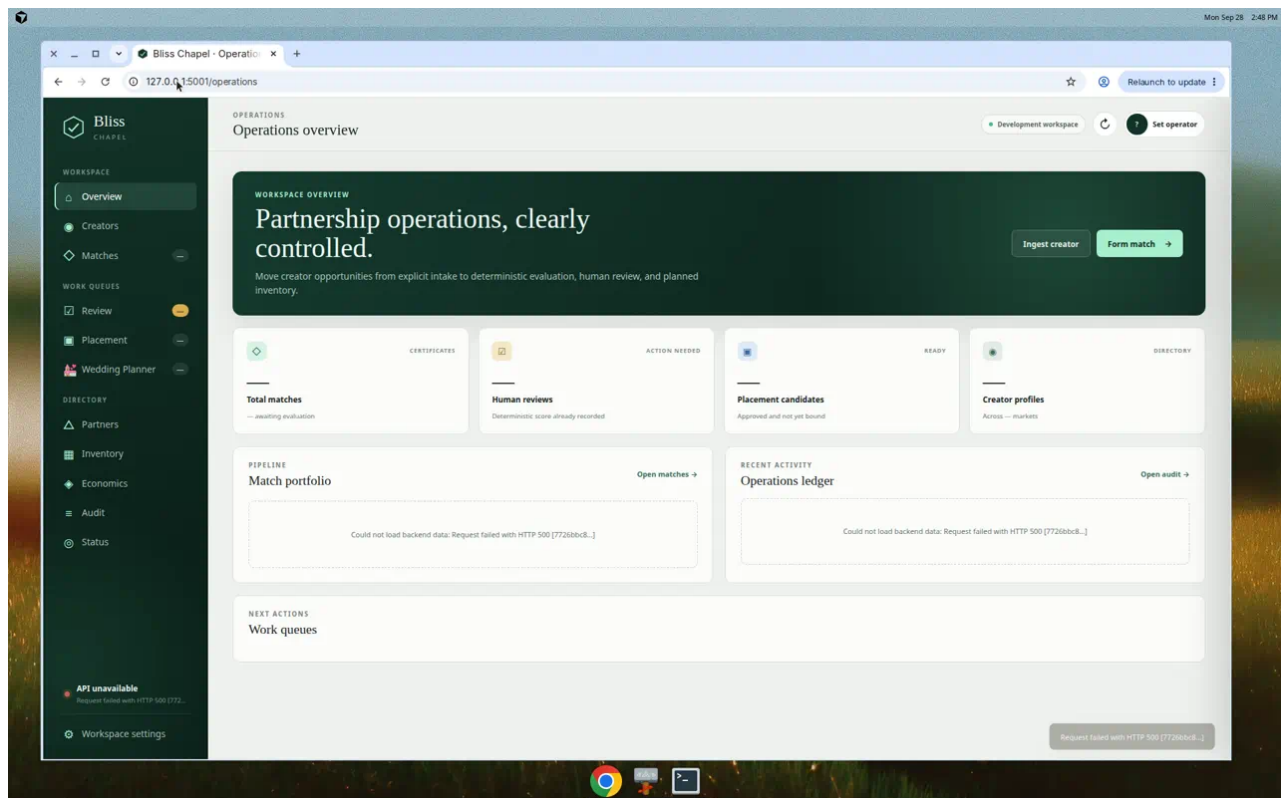


Figure 2. Operations remained usable and reported the database failure inside the panels.

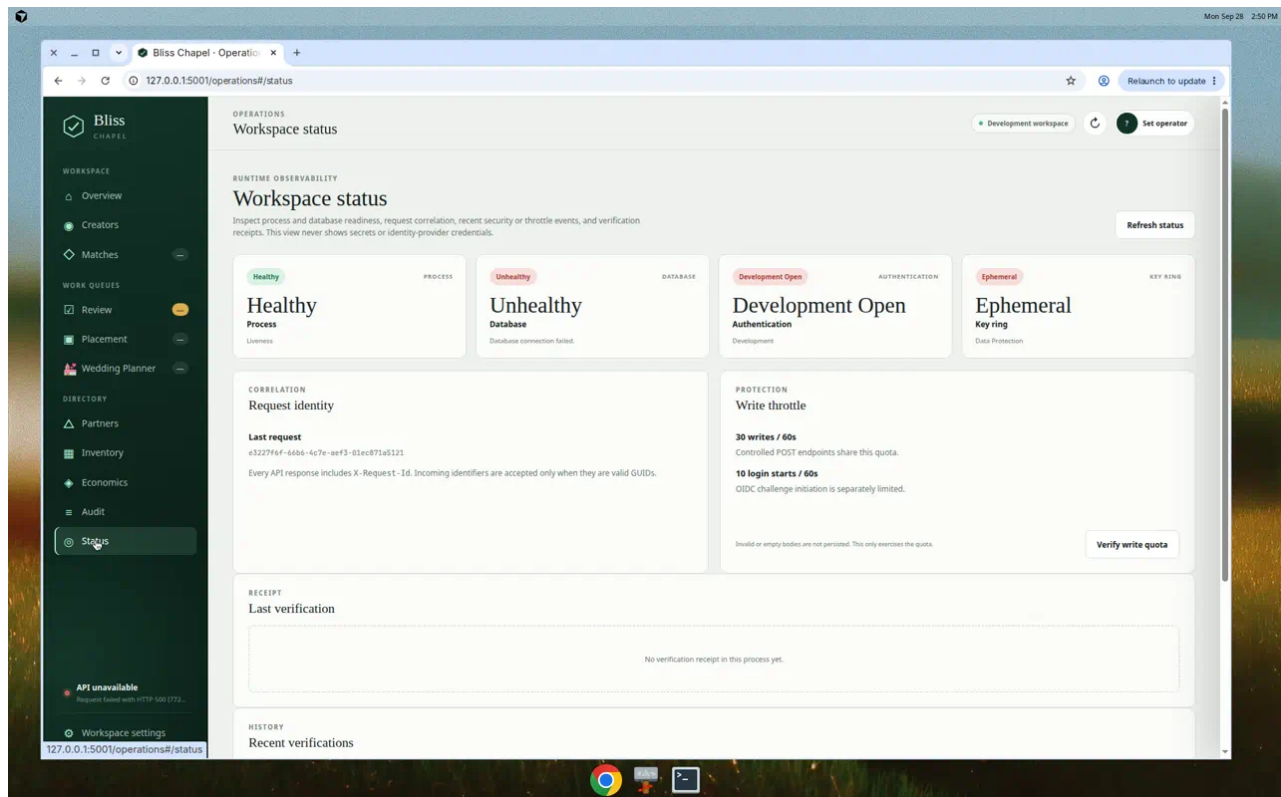


Figure 3. Status showed Process Healthy and Database Unhealthy. The key ring was still Ephemeral.

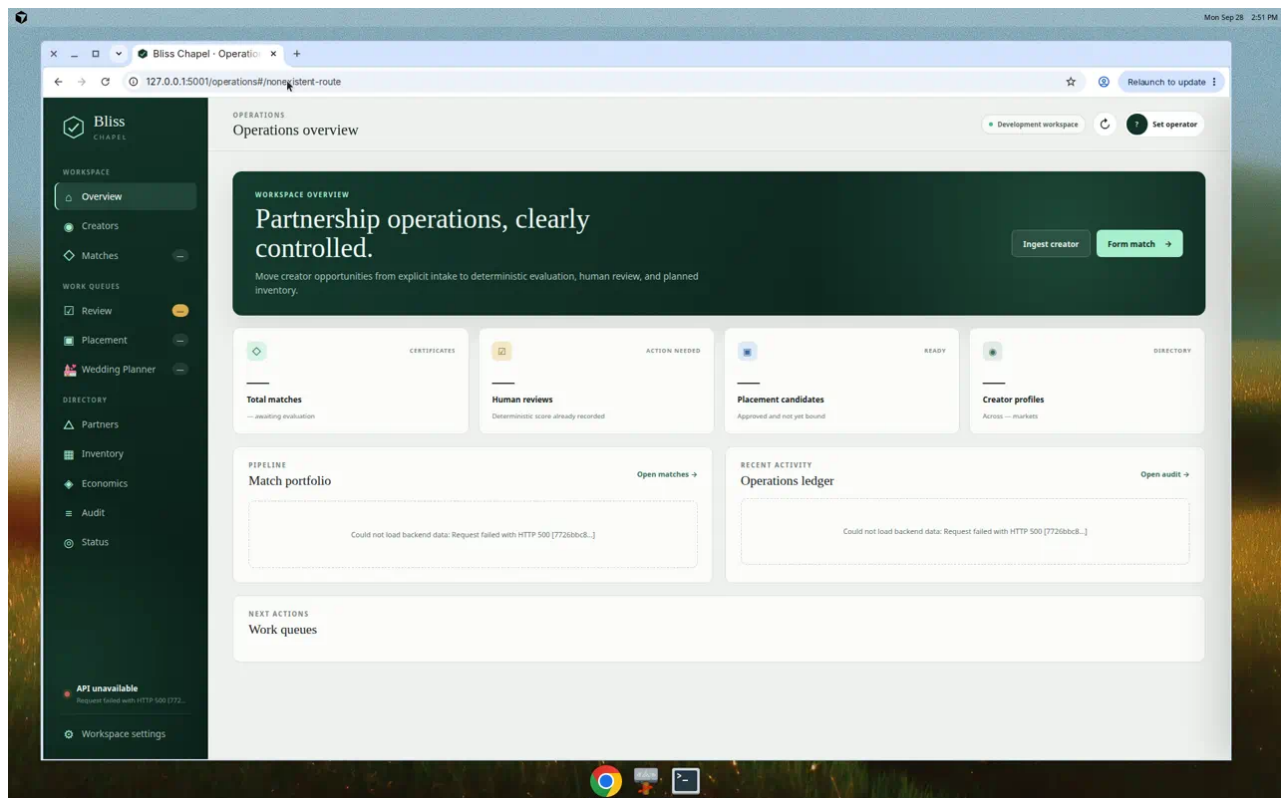


Figure 4. An unknown operations route fell back to the overview.

PostgreSQL and OIDC acceptance

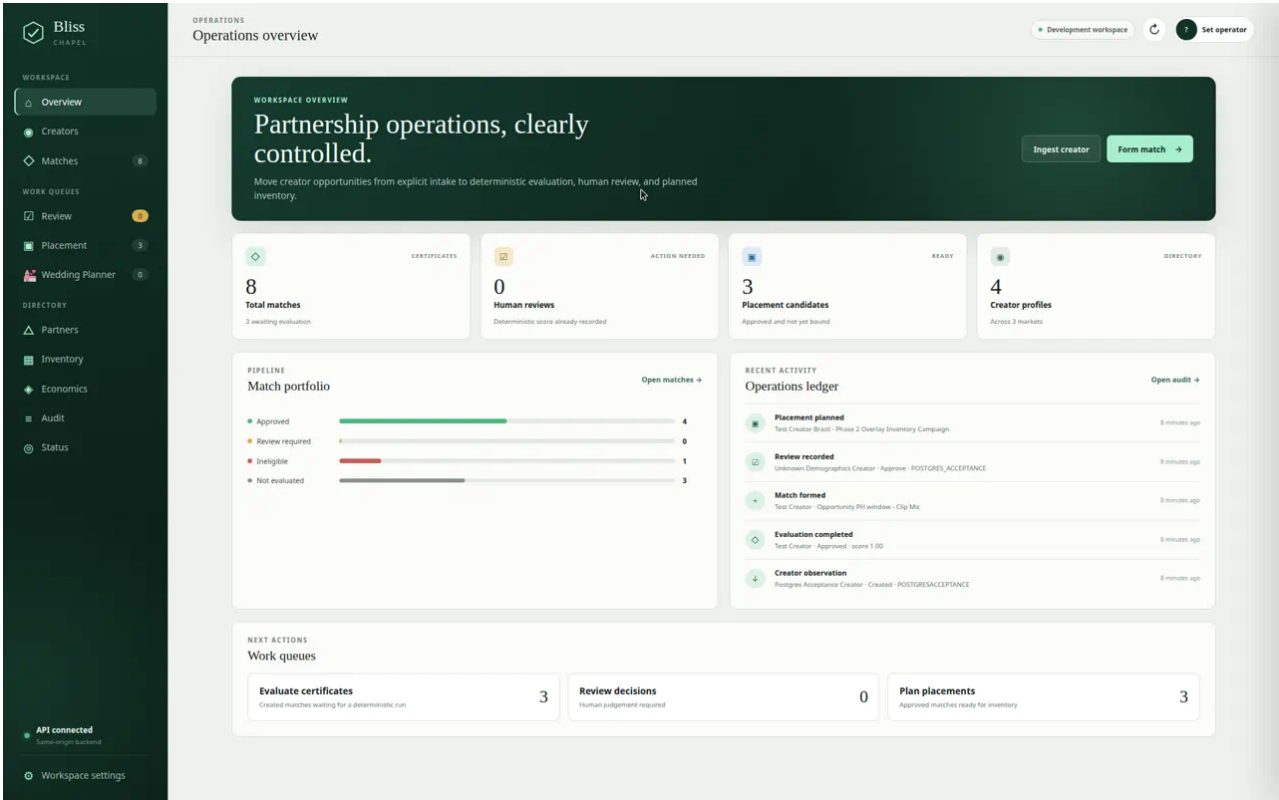


Figure 5. PostgreSQL-backed overview: API connected, 8 matches, 4 creators, and the workflow ledger.

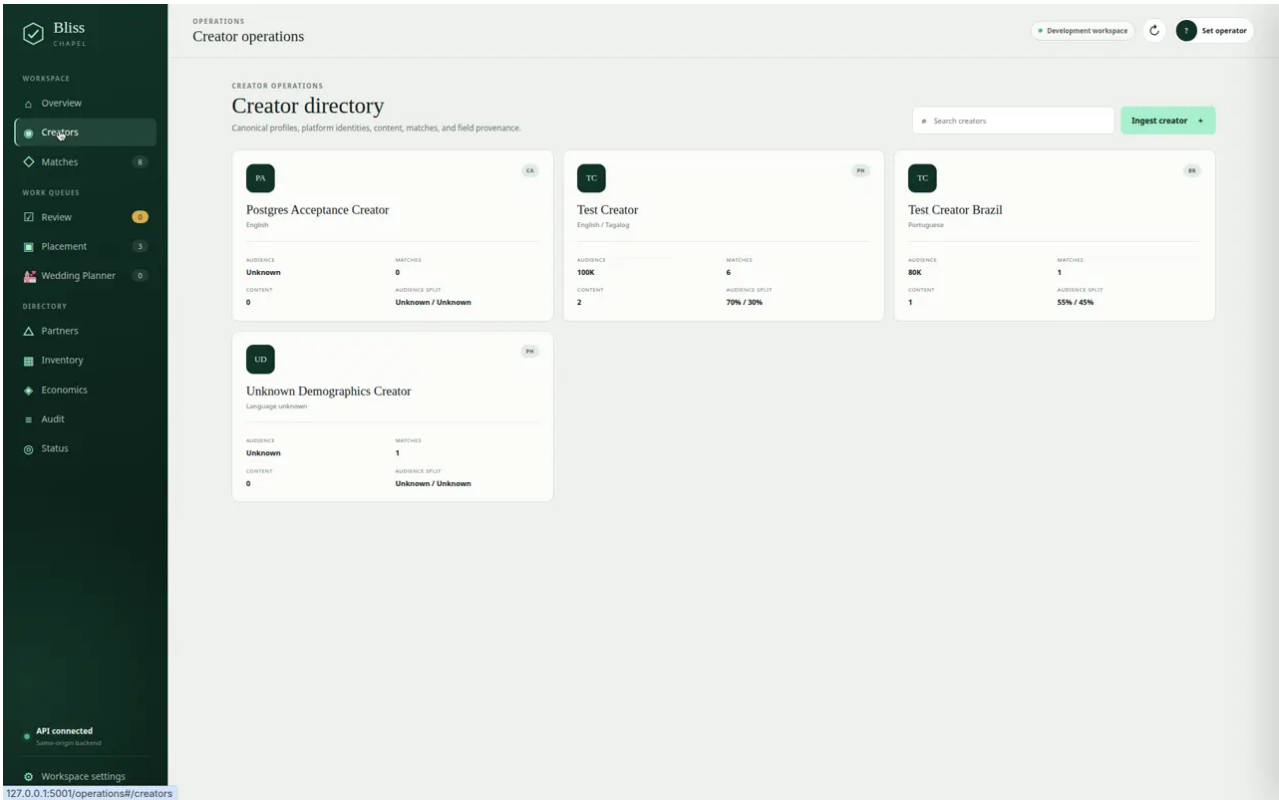


Figure 6. Creator directory, including Postgres Acceptance Creator created through the API.

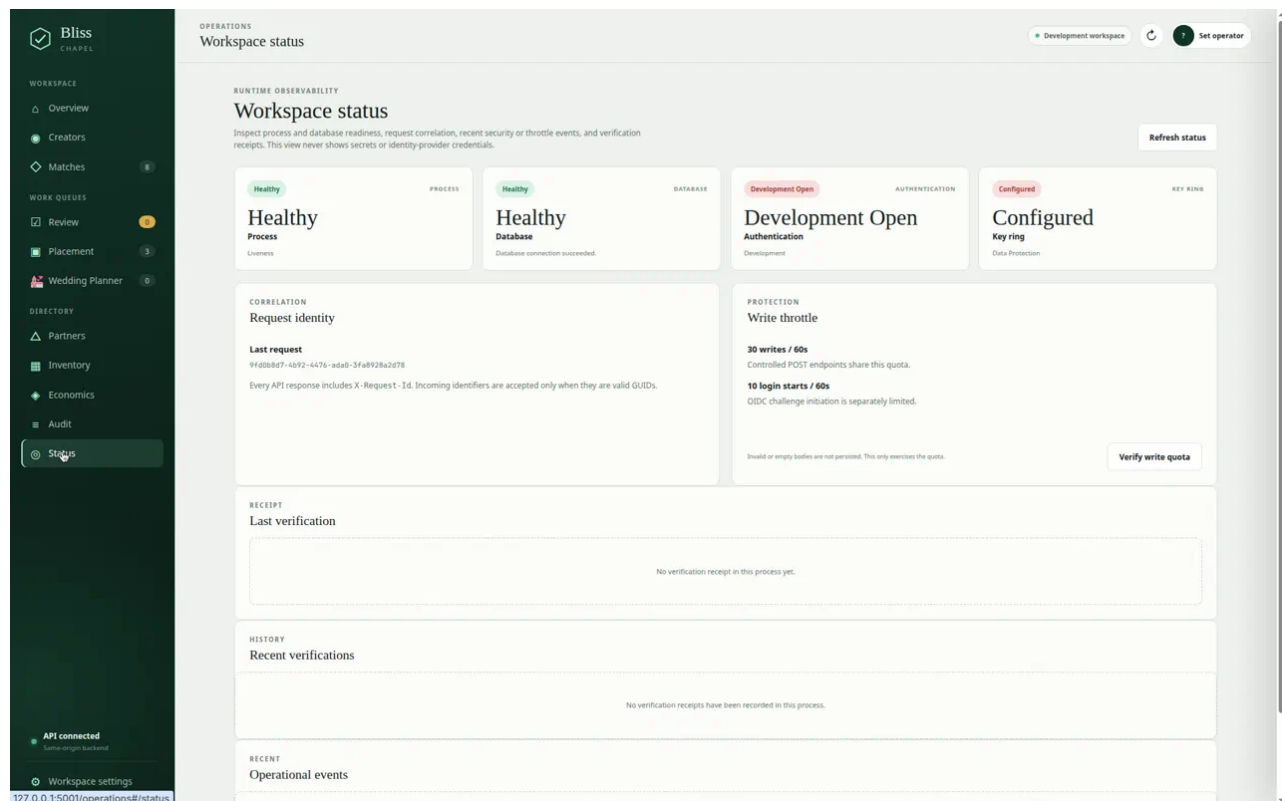


Figure 7. Development workspace: Database Healthy and Key ring Configured. Authentication remains Development open in that mode.

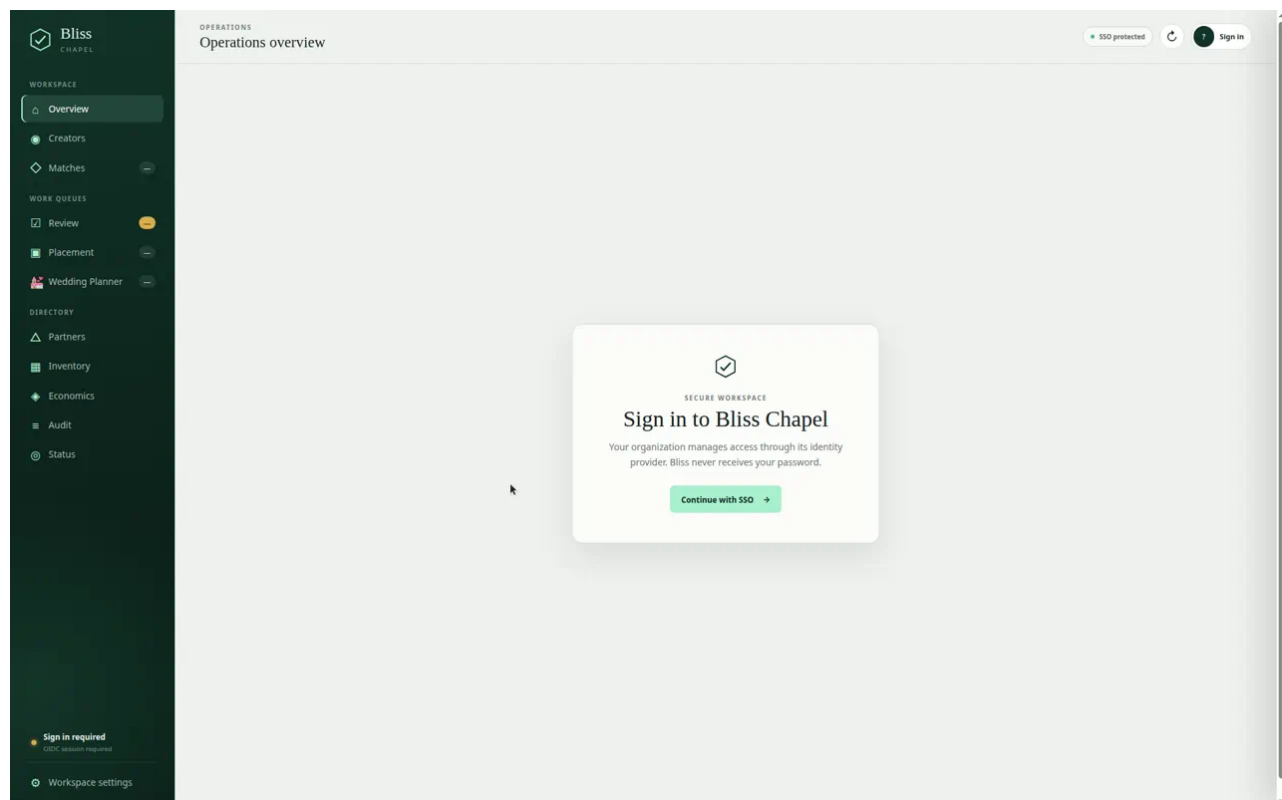


Figure 8. Production acceptance workspace required an OIDC session before showing data.

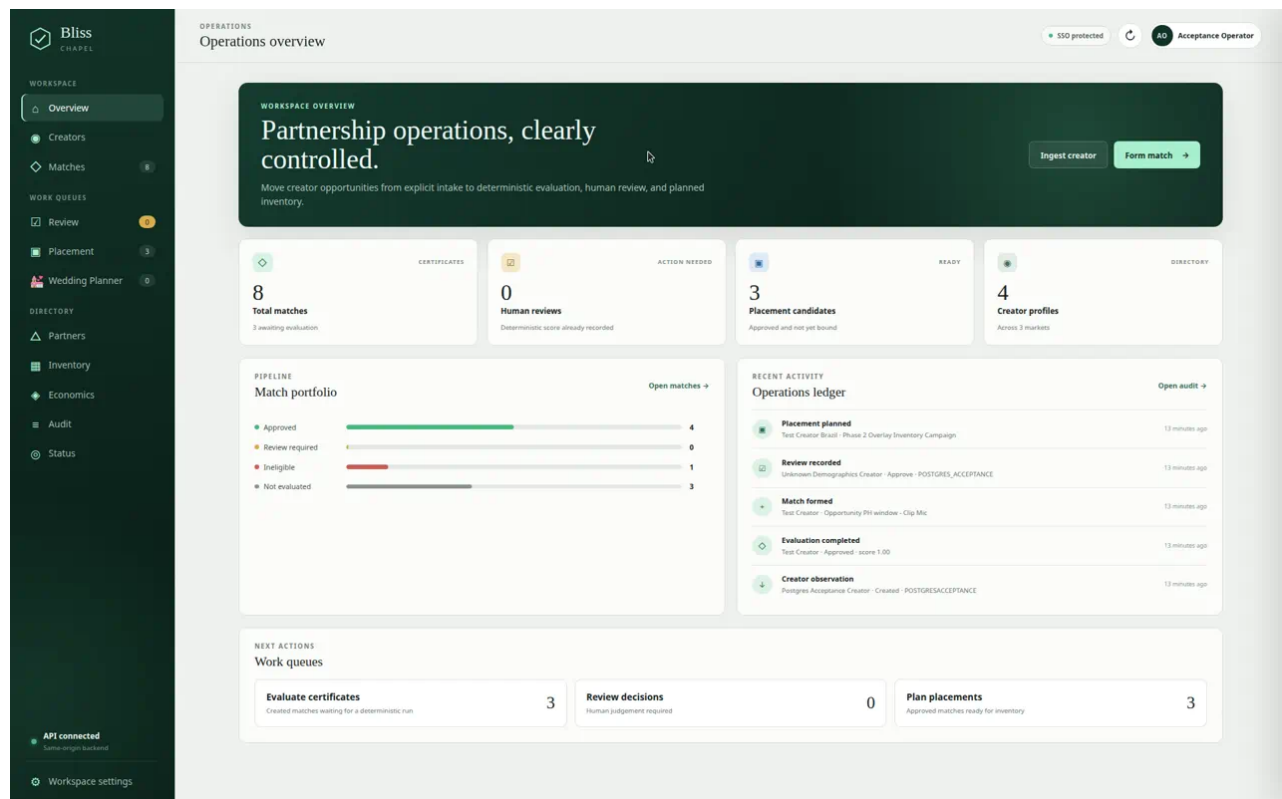


Figure 9. After SSO, the same PostgreSQL records loaded for Acceptance Operator.

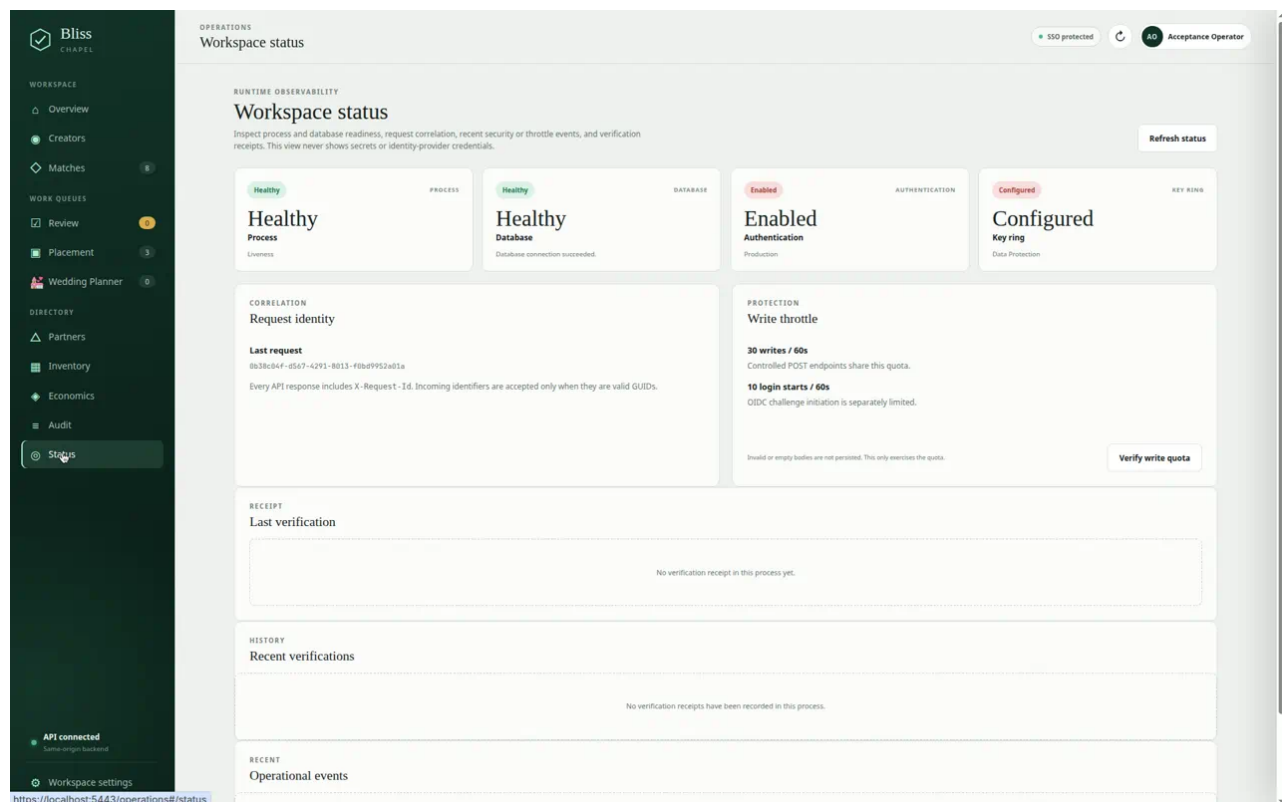


Figure 10. Acceptance status: Process Healthy, Database Healthy, Authentication Enabled, Key ring Configured.

6. Limits

- Load figures are single-host development measurements, not a production capacity limit.

- The OIDC provider was a local non-production acceptance authority, not an organizational identity provider.
- Role authorization requires the roles claim in the ID token. A userinfo-only role was not applied.
- The persisted data-protection key is not encrypted at rest.
- The development process intentionally leaves authentication disabled. The Production acceptance process is the authenticated result.
- No repository files were changed.

Recorded browser evidence is saved as bliss-postgres-oidc-acceptance.mp4. The earlier database-unavailable recording is bliss-system-negative-regression-demo.mp4.